

Thesis progress report

LLM-Guided Neural Network Compilation for Real-time UAV Edge Inference

Atul Kumar Gupta · 25MCSS06

A page that runs the loop

Compile, check, and measure were separate commands. They are the same functions here, called from one page on this machine.

The page shows the host, the graph notes, the plan and its attempts, the ranking, a table across the seven models, and a fresh timing of the chosen file. The operator picks a host and a model id. If the key is missing, the page says key not set. It does not ask for the key.

The seven ONNX files are stored in experiments/models.

The page listens only on this machine, at 127.0.0.1. One job runs at a time. If the console process is down, the buttons become usable again and the bar says the console is not reachable. Analyze, plan, optimize, export, and a remeasure of the chosen file all call the same Python functions as the command line.

A markdown table was the first way to show the work. It hid the attempts. A refused step, a rate limit, and a faster plan that failed the check all lived in JSON that a reader would not open. The page had to show those, or the demonstration would slide back to the fastest number. An early version also locked every button after one failed request, which looked like a dead program. The lock now clears, and the bar says why.

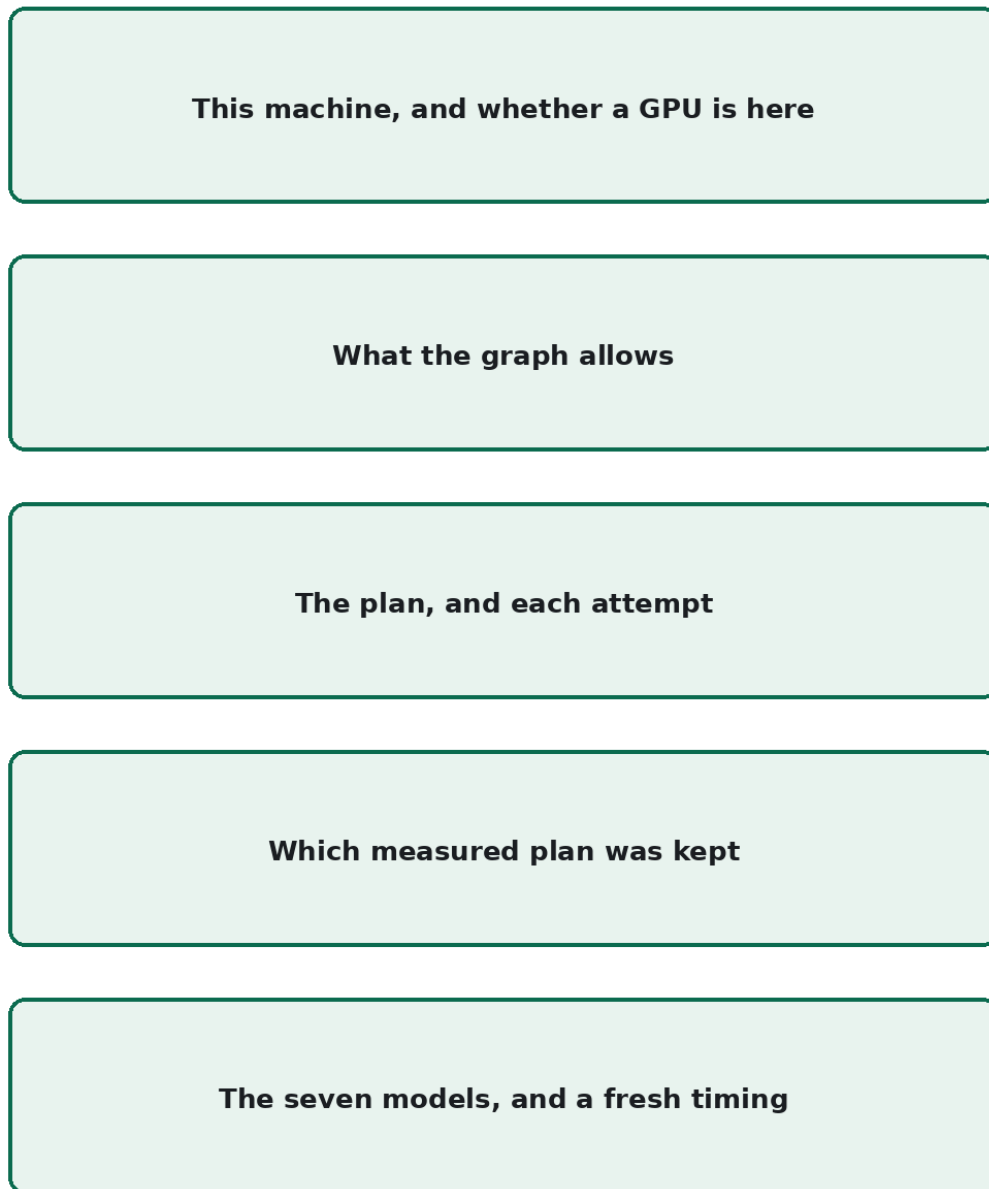


Figure 1. The page is those five bands, in that order.

The loop can be run and shown without reading raw logs. What the thesis is allowed to claim is a separate statement.